

CIS 21031 – PRACTICAL FOR PLATFORM TECHNOLOGIES
DEPARTMENT OF ICT
FACULTY OF TECHNOLOGY
SOUTH EASTERN UNIVERSITY OF SRILANKA

Labsheet: 05

Date: 05.12.2023

Title: Programming Shell Scripts- III

Aim: Getting practice to write a shell script for Conditional Execution (Decision Making)

Tasks:

- Working with numerical comparisons in shell script
- Working with string comparisons in shell script
- Working with file test operators in shell script

Exercise 1: Numerical Comparison

For Numerical Comparison

Operator	Meaning
-eq	Is equal to
-ne	Is not equal to
-gt	Is greater than
-lt	Is less than
-ge	Is greater or equal
-le	Is less than or equal

1. Write a shell script to check whether two numbers are equal or not.
2. Write a shell script to find to greatest of two numbers.
 - a. Get the two numbers from the user.
 - b. Display which of two numbers is greater.

3. Write a shell script to check whether a number is positive or negative.
 - a. If the given number is positive, you should print "<<Number>> is a positive number".
 - b. Else you should print "<<Number>> is a negative number".
 - c. Else print "Neither positive Nor negative"
4. Write a shell script to determine whether a given number is either odd or even.
 - a. If the given number is odd, you should print "<<Number>> is an odd number"
 - b. Else you should print "<<Number>> is an even number"
5. Write a Bash script to determine whether a student has passed or failed based on their exam marks.
 - Prompt the user to input the student's marks
 - Display an appropriate message indicating whether the student has passed or failed.

Exercise 2: String Comparison

For String Comparison Use:

Operator	Meaning
String1 == String2	String1 is equal to String2
String1 != String2	String1 is not equal to String2
-n string1	string1 is NOT NULL and does exist
-z string1	string1 is NULL and does exist

1. Get the two strings from the user compare two strings are,
 - a. Two strings are Equal or Not
 - b. Check if the string is empty.

Exercise 3: File test Operator

For File Test Operators

Test	Meaning
-s file	Check if file has size greater than 0.
-e file	Check if file exists
-r file	Check if file is readable
-w file	Check if file is writable.
-x file	Check if file is executable.
-f file	Check if file is an ordinary file.
-d file	Check if file is a directory

1. Create a shell that can check the given file test operators for an existing file in your current working directory.
 - Check for read, write, and execute permission.
 - Check whether the file is directory or not.
 - Check whether the file exists or not.
2. Write a shell script to check if a directory exists or not.
 - a. Check whether (/Desktop/<<Your Registration Number>>) directory is there in your machine.
 - b. If it is there, Print “Machine has a directory”
 - c. If it is not available, then create that directory.

Practice Question:

1. Write a shell script to find the greatest of three numbers.
2. Write a shell script to check whether two strings are equal or not.
 - a. Get the two strings from the user.
 - b. Display two strings are equal or not.
3. Write a shell script to check if a file exists or not.

Assume that you are going to check whether the file named ***file_ICTXXX.sh*** exists or not by running the shell script file.

- a. Create a folder in your current working directory.
- b. Navigate to the newly created folder and create the file.
- c. To check whether the file exists or not write the script by using test operators along with if clause.
- d. Run the script and get the output.
- e. Now try to remove the file and rerun the script.